



Effects of drying variables on the characteristic of the hot air drying for gastrodia elata: Experiments and multi-variable model



K. Li ^{a, d}, Y. Zhang ^b, Y.F. Wang ^b, W. El-Kolaly ^{c, e}, M. Gao ^b, W. Sun ^a, M. Li ^{b, *}

^a Yunnan Normal University, School of Physics and Electronic Information, Juxian Str, 768, Kunming, China

^b Yunnan Normal University, Solar Energy Research Institute, Juxian Str, 768, Kunming, China

^c Kunming University of Science and Technology, Faculty of Metallurgical and Energy Engineering, Jingming Str, 727, Kunming, China

^d Power Construction Group of China & Hainan Electric Power Design Research Institute Company Limited, Guoxing Str, 268, Haikou, China

^e Agricultural Engineering Research Institute (AENRI), Agricultural Research Center (ARC), Ministry of Agriculture, 256Dokki, Giza, Egypt

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ABSTRACT

Energy saving and efficient drying model play important role in the extraction of high quality Chinese medicinal materials. In this work, the gastrodia elata was subjected to hot air drying of revealing the drying characteristics with the medicinal indicator of gastrodin ingredient. The multiple drying variables of drying temperature, air velocity, slice thickness and relative humidity were taken into account. Based on the experimental results, an optimized multi-variable model was established to predict the characteristics of drying the sliced gastrodia elata with hot air. The experimental results show that the optimal model of the Page model enables to predict the water loss process of the sliced gastrodia elata under multiple drying variables. The optimal drying variables were determined based on the maximum gastrodin content, of which the temperature was 60 °C, air velocity was 3.5 m/s, slice thickness was 0.4 cm and relative humidity was 25 %. Meanwhile, the drying quality was improved efficiently. The effective diffusion coefficient of gastrodia elata in the hot air drying ranged from $2.254 \times 10^{-9} \text{ m}^2/\text{s}$ to $3.241 \times 10^{-8} \text{ m}^2/\text{s}$, and the dry activation energy was 34.16 kJ/mol.

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1. Introduction

Traditional Chinese medicines play a vital role in the medical and health care, which are mainly comes from fresh plants. Drying technology is essential to prepare the Chinese medicines, in order to remove high water content in the fresh plants and thereby achieve safe, healthy, high-quality and long-term storage. However, most existing drying technologies relied on the work experience from repeat experiments to obtain key parameters to one specific Chinese herbal medicine drying. The theoretical model of the drying mechanism of traditional Chinese medicine has still to be established, since the basic drying process and characteristics remains to be revealed in-depth. It results in a lot of blindness in choosing drying technologies and equipment, which inhibits the promotion and application of appropriate drying process. The state-of-the-art drying technologies mainly include hot air drying, vacuum freeze drying, infrared drying, and microwave vacuum

drying [1–4]. Compared to the hot air drying, the other drying methods have the disadvantages of complicated equipment, high investment, large energy consumption, and insufficient drying amount. Thus, the hot air drying is still the main technology to prepare the Chinese herb medicine [5–7].

The drying effect depends on the drying parameters, which is usually characterized by the main medicinal ingredient of a specific plant. Among these medical plants, gastrodia elata is one kind of perennial rhizome herb of orchidaceous, which is a valuable traditional Chinese medicine with high medicinal value and edible value [8]. Gastrodia elata lives underground during its life cycle except for florescence. The whole plant is free of chlorophyll, which is the unique botanical character of it. It grows to the height of 30 cm–150 cm and the stems are vertical, cylindrical and yellowish red. Its inflorescence is fringy raceme, golden colored and 10 cm–30 cm long. The leaves are scale-like, membranous and about 1 cm–2 cm long. The rhizomes are pedantic, oblong, about 10 cm long, and 3 cm–4.5 cm in diameter. The main medicinal ingredient indicator of the gastrodia elata is the gastrodin that has a wide range of clinical applications in western medicine [8,9]. Thus, the content and quality of gastrodin should be explored during the

* Corresponding author.
E-mail address: lmillyd@126.com (M. Li).

drying process and under the multiple drying parameters. It should be mentioned that the drying quality of other products was also mainly dominated by the drying parameters. Hazervazifeh et al. studied the drying methods for processing apple slices, of which the energy conservation approach was identified under different hot air temperatures, inlet air velocities, as well as microwave power levels [10]. Mohammadi et al. used a hot-air dryer to dry kiwifruit slices at three different temperatures, where two different models were applied to determine mass transfer and activation energy [11]. Motevali et al. observed that the increase of temperature and air velocity lead to the decline of energy consumption in drying mushroom [12].

Recently, researches on drying the gastrodia elata have been increasing, but the state-of-the-art remains in specific parameter analysis rather than revealing the dynamic change of its water content for the higher gastrodin. Tian et al., illustrated that the content of gastrodin after steaming the gastrodia elata was higher than that of the cooked and untreated fresh gastrodia elata [13–15]. Qu et al. identified the main and secondary factors that influenced the process of the hot air drying after steaming the gastrodia elata [16]. The results showed that the optimum factors corresponded to the temperature of 60 °C, cooking time of 16 min, humidity of 30 % and spreading density of 5 kg/m².

Even though there are some prediction models for plants drying, the applicability and generality for the gastrodia elata remains to be developed. Chen et al., studied the effects of temperature (from 40 °C to 80 °C) and slice radius (from 0.8 cm to 1.2 cm) on the drying characteristics of gastrodia elata by infrared drying method [17]. Then, six models were compared to verify suitable formulas for predicting the dehydration process of gastrodia elata in drying environment with the temperature and slice radius, respectively. The results illustrated that the smaller radius with higher temperature resulted in the fastest drying rate for infrared drying of gastrodia elata, while the better drying temperature ranged between 50 °C and 60 °C. Besides, the Page drying model showed the best accuracy to predict the drying and water loss process of gastrodia elata. Wang and Xu studied the influence of temperature (55 °C–90 °C) and air velocity (0.6 m/s – 3.0 m/s) on the drying characteristics of gastrodia elata using the hot air drying. They established the Page model to predict the water loss process under the different temperature and air velocity environments [18]. It was found that the combination of three temperatures and air velocities at (65 °C, 1.0 m/s), (75 °C, 0.6 m/s) and (75 °C, 1 m/s) gave the higher drying efficiency, lower energy consumption, and the better product quality. Hossaind compared three prediction models of the dehydration process of green pepper with a drying temperature of 40 °C–65 °C, air velocity of 0.1 m/s – 1.0 m/s and relative humidity of 10 %–60 %, where thin layer hot air drying method was applied [19]. It has been observed that the Page model was better than the single exponential equation to describe the drying process of thin layer drying of chili. However, the model was just for vegetables. In general, mathematical models were still insufficient to predict the dynamic change of the water content with the multiple drying variables of the gastrodia elata, which enables to optimize the process parameters for maintaining higher content of the gastrodin.

This study aims to analyze the effects of drying temperature, air velocity, relative humidity and slice thickness on gastrodia elata drying characteristics and gastrodin content with the hot air drying technology. Meanwhile, the heat and moisture transfer model was established under above multiple drying parameters. Then, the optimum parameters for drying gastrodia elata with the maximum gastrodin content were determined to improve the drying quality at a short time. Meanwhile, a heat and moisture transfer model was established and verified by the experimental results to predict the

water loss process of gastrodia elata tablets under different drying conditions. The research is beneficial to provide a universal characteristic drying rule for the drying of rhizome Chinese medicinal materials, which is expected to give a reference on the industrial drying of rhizome Chinese medicinal materials with energy saving and high efficiency.

2. Materials and methods

2.1. Hot air drying system for the Chinese medicinal materials

Fig. 1 illustrates the design and photograph of the hot air drying system, which enables to provide drying environment with different hot air temperature, air velocity and relative humidity for different materials to facilitate the appropriate water evaporation.

The dryer system consists of the heat pump cycle and the dry air cycle. In the heat pump cycle, the liquid refrigerant was throttled by the expansion valve, and then became a low-temperature and low-pressure liquid entering the evaporator. During this process, the refrigerant was vaporized by absorbing heat from the outdoor air and recovering the heat of the hot steam. Subsequently, the low-temperature and low-pressure gaseous refrigerant was compressed by the compressor. And then, it entered the condenser to transfer the heat to the air in the dryer. In order to control the change of the temperature and relative humidity of the air in the dryer, the refrigerant flow was controlled by adjusting the valve located at the condenser inlet. And the hot air with the desired air temperature was blown into the dryer by a fan. Besides, an electric heater was installed at the dryer inlet as an auxiliary heater in case the air temperature was below the desired temperature.

In the hot air cycle, the pre-heated air was blown into the drying box by a fan. As a result, the heat of air was transferred to the gastrodia elata. In the process, the moisture in the gastrodia elata was vaporized and then taken out of the drying oven by a return air fan.

All factors of the dryer were controlled by the controller unit, of which the air temperature was controlled from 0 °C to 100 °C, relative humidity from 20 % to 98 %, and air velocity from 0 m/s to 5 m/s. The specifications of the instruments were shown in Table 1.

2.2. Sample preparation and measurements

The first-class gastrodia elata (weight 200 g–250 g) were washed, peeled, and steamed. The optimal steaming time was 20 min with the temperature of 105 °C to ensure the surface transparent, bright and interior color without a white heart [15]. The steamed gastrodia elata was sliced with different thicknesses of 0.2 cm, 0.4 cm, 0.6 cm and 0.8 cm, respectively.

The method in GB5009.3–2010 was used to test the initial moisture content of gastrodia elata by the halogen moisture meter [20]. Each sample was tested 5 times. The initial moisture content of gastrodia elata ranges from 80 % to 84%, with an average of 82 %. Moreover, the active medicinal ingredient of gastrodia elata was tested by the HPLC method of the Chinese Pharmacopoeia using high-performance liquid chromatography [8].

The sliced gastrodia elata was dried in a single layer on four square mesh shelves with area of 0.64 m², where five shelf was placed inside the hot air dryer oven. The drying parameter of gastrodia elata slices as shown in Table 2. The drying temperatures were kept at 45 °C, 50 °C, 55 °C, 60 °C and 65 °C, respectively. The corresponding measured relative humidity at each temperature was 20 %, 25 %, 30 % and 35 %, respectively. The air velocities were measured at 1.5 m/s, 2.0 m/s, 2.5 m/s, 3.0 m/s, 3.5 m/s and 4.0 m/s, respectively. The drying was continued until there were no noticeable changes in the mass of the slices as shown in Fig. 2. Each

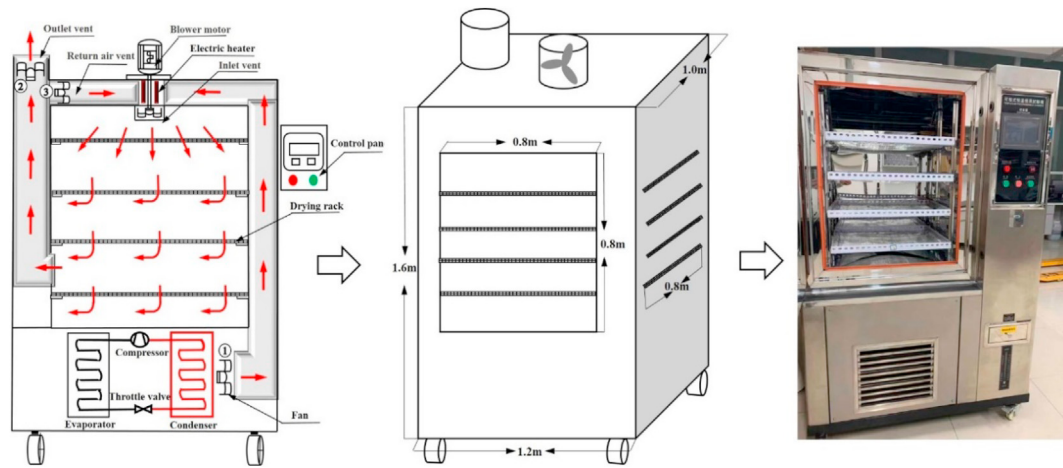


Fig. 1. Hot air drying system.

Table 1
Specifications of the instruments.

Instrument name	Measurement range	Accuracy
Constant temperature and humidity drying oven	Temperature: 0–100 °C Relative humidity: 5–98 % Wind speed: 0–10 m/s Geometric dimensions of the dryer	±0.5 °C ±1% RH ±0.05 m/s 0.512m ³
Halogen moisture meter	Electric heater Compressor	1 kW 2.63 kW
Slicer	Heating temperature: 40–200 °C	Weighing accuracy: 0.001 g
Electronic balance	Weighing: 0–120 g Slice thickness: 0–18 mm	Weighing accuracy: 0.01 g
High performance liquid chromatography (NextEra LC-20AD)	Flow: 0–10 mL/min Drain pressure: 0–40 MPa Wavelength: 190–700 nm	Flow: 2 µL/min Wavelength: 1 nm

Table 2
Gastrodia elata slices at different drying parameter.

	Temperature °C	Air velocity m/s	Slice thickness cm	Relative humidity %
Temperature	45	2.0	0.4	20
	50	2.0	0.4	20
	55	2.0	0.4	20
	60	2.0	0.4	20
	65	2.0	0.4	20
Air velocity	60	1.5	0.4	20
	60	2.0	0.4	20
	60	2.5	0.4	20
	60	3.0	0.4	20
	60	3.5	0.4	20
Slice thickness	60	4.0	0.4	20
	60	2.0	0.2	20
	60	2.0	0.4	20
	60	2.0	0.6	20
	60	2.0	0.8	20
Relative humidity	60	2.0	0.4	20
	60	2.0	0.4	25
	60	2.0	0.4	30
	60	2.0	0.4	35

drying process was completed in triplicate and the mean value was used in the analysis. The measurement was performed every 1 h.

2.3. Determination of drying parameters

The drying rate, wet base moisture content, dry base moisture content and water ratio of the gastrodia elata slices at the different

time are calculated as follows [21–23].

The drying rate (D_s) of materials at any time is calculated according to Eq. (1):

$$D_s = \frac{M_t - M_{t+\Delta t}}{\Delta t} \quad (1)$$

The moisture ratio (M_c) of the material at any drying time is



Fig. 2. *Gastrodia elata* slices at different drying parameter (a) Fresh sliced *gastrodia.elata*; [Sliced *gastrodia.elata* under the different drying condition (b) Temperature 70 °C, air velocity 2.0 m/s, relative humidity 20 %, slice thickness 0.4 cm. (c) Temperature 65 °C, air velocity 2.0 m/s, relative humidity 20 %, slice thickness 0.4 cm. (d) Temperature 60 °C, air velocity 2.0 m/s, relative humidity 20 %, slice thickness 0.4 cm. (e) Temperature 60 °C, air velocity 3.5 m/s, relative humidity 25 %, slice thickness 0.4 cm. (f) Temperature 60 °C, air velocity 3.5 m/s, relative humidity 25 %, slice thickness 0.4 cm. (g) Temperature 60 °C, air velocity 3.5 m/s, relative humidity 25 %, slice thickness 0.2 cm. (h) Temperature 60 °C, air velocity 3.5 m/s, relative humidity 25 %, slice thickness 0.4 cm].

calculated by:

$$M_c = \frac{C_t}{W_g + C_t} \quad (2)$$

The dry basis moisture ratio (M_t) of the material at any drying time is defined as:

$$M_t = \frac{C_t}{W_g} \quad (3)$$

MR refers to the remaining moisture content of the material under certain drying conditions. It can reflect the speed of drying under this drying condition. The material moisture ratio MR is calculated by:

$$MR = \frac{M_t}{M_0} \quad (4)$$

3. Results and discussions

3.1. Effect of temperature on the characteristics of *gastrodia elata*

Fig. 3 shows the effect of temperatures on the sliced *gastrodia elata* drying rate, water loss and gastrodin content at constant relative humidity 20 %, slice thickness 0.4 cm, and air velocity 2.0 m/s in the hot air drying.

Fig. 3(a) shows that the gastrodin content increased first and then decreased with the increase of temperature. The gastrodin content increased linearly between 45 °C and 60 °C, the content of gastrodin was 0.503 % at 60 °C, which was 2.5 times higher than the standard of the Chinese pharmacopoeia for the *gastrodia elata* [8].

The gastrodin content decreased, when the temperature was above 60 °C. It can be inferred that the hot air temperature should be lower than 60 °C, it is benefit for the gastrodin-generating enzyme to redox the internal components to gastrodin. The reason is that when the temperature is above 60 °C, the balance of redox level would be broken. It leads to the gradual decrease of the activity of the gastrodin generating enzyme, while the activity of gastrodin degrading enzyme gradually increases.

As a result, the content of gastrodin production decreases. The results reflect that the higher temperature caused a mutual transformation of internal components so that the gastrodin content decreases, which are consistent with the conclusion from Ji et al [24]. Thus, the temperature has a great effect on the gastrodin content, of which the optimum temperature is 60 °C with the conditions abovementioned.

Fig. 3(b) shows the drying process of the *gastrodia elata* has two stages with the increase of temperature, namely the speed-increase stage and speed-reduction stage. The reason is as follows. The speed-increase stage is the pre-heating stage of the *gastrodia elata*, of which the free water on the surface of *gastrodia elata* was evaporated gradually. The free water evaporation gradually turned into combined water evaporation at the speed reduction stage, while the stage is the main drying process of *gastrodia elata*. On the other hand, the preheating period and drying period were gradually shortened with the increase of temperature. The drying period at 65 °C was 1.75 time shorter than that at 45 °C. The reason is that with the increase of air temperature, the drying time consumption is reduced. This is attributed to the decrease of the evaporation pressure, which reduces the moisture evaporation resistance from materials and accelerate the moisture evaporation. The conclusion was supported by Hazervazifeh et al [10].

Fig. 3(c) shows the slope of the drying curve under different

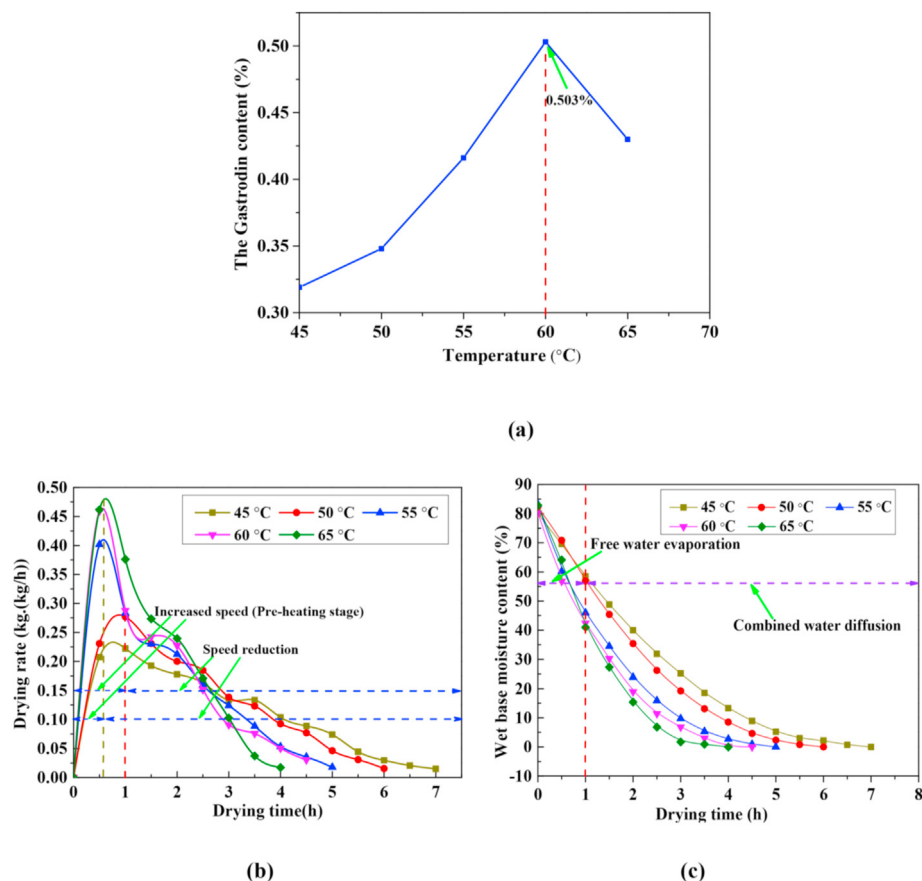


Fig. 3. Effect of temperature on the: (a) gastrodin content; (b) drying rate; (c) water loss rate.

temperatures. It indicates that the water content rapidly reduces with the increase of temperature. Thus, the temperature has great impact on the drying process of water loss. The large distance between the drying curves was between 50 °C and 55 °C and the wet base moisture content at 55 °C was significantly lower than that at 50 °C. Therefore, the water content rapidly reduced once the temperature was above 50 °C. But when the temperature is too high, the color of gastrodia elata would be affected. Specifically, the color of gastrodia elata was yellow at 65 °C, and it becomes white at 60 °C. Therefore, the optimum drying rate corresponds to the temperature of 60 °C, where the gastrodia elata presents good quality with the highest content of gastrodin, as well as white color.

3.2. Effect of air velocity on the characteristics of gastrodia elata

Fig. 4 shows the effect of air velocity on the sliced gastrodia elata drying rate, water loss and gastrodin content under the drying condition of relative humidity 20 %, slice thickness 0.4 cm, and temperature 60 °C in the hot air drying.

Fig. 4(a) shows that the gastrodin content rapidly increased from 0.303 % to 0.503 % with the air velocity increasing from 1.5 m/s to 2.0 m/s. When the air velocity was increased from 2.0 m/s to 3.5 m/s, the gastrodin content increased slowly from 0.503 % to 0.53 %, while the gastrodin content was 2.65 times higher than the standard of the Chinese pharmacopoeia for the gastrodia elata at the air velocity of 3.5 m/s [8]. It indicates that the activity of gastrodin enzymes is highest at the air velocity of 2.0 m/s. When the air velocity is above 2.0 m/s, the activity of gastrodin degrading enzymes gradually increases, and the activity of gastrodin generating enzymes decreases rapidly. Thus, the air velocity has little

effect on the gastrodin content.

The drying rate curve went steeper as the air velocity increased from 1.5 m/s to 4.0 m/s under the pre-heating time at 30 min. Correspondingly, the drying rate inclined from 0.213 kg/(kg/h) to 0.45 kg/(kg/h), as shown in Fig. 4(b). This is caused by the reduction of the moisture evaporation pressure and the moisture diffusion resistance in materials as the decrease of air velocity. The findings were consistent with the previous researches [25,26]. The distance between air velocity curves at 1.5 m/s and 2.0 m/s was large, but the air velocity above 2.0 m/s made little changes in the slope distance of the drying curve, as shown in Fig. 4(c). Therefore, the drying water loss process of gastrodia elata slices is insensitive to the air velocity. In addition, the air velocity of 3.5 m/s and 4.0 m/s have the same effect on the safe moisture content. Generally, the air velocity of 3.5 m/s enables to achieve the optimum drying rate and the highest gastrodin content.

3.3. Effect of slice thickness on the characteristics of gastrodia elata

Fig. 5 shows the effect of gastrodia elata slice thickness on drying rate, water loss and gastrodin content at constant relative humidity 20 %, air velocity 3.5 m/s, and temperature 60 °C in the hot air drying.

As the slice thickness increases from 0.2 cm to 0.4 cm, the gastrodin content also increases from 0.405 % to 0.503 %. Whereas, as the slice thickness increases from 0.6 cm to 0.8 cm, the gastrodin content decreases from 0.49 % to 0.416 %, as shown in Fig. 5(a). The reason is that the gastrodin is easy soluble in water [8]. The thicker the slice, the longer the distance of water transferring from the material, and the less water transfer at the same time, resulting in

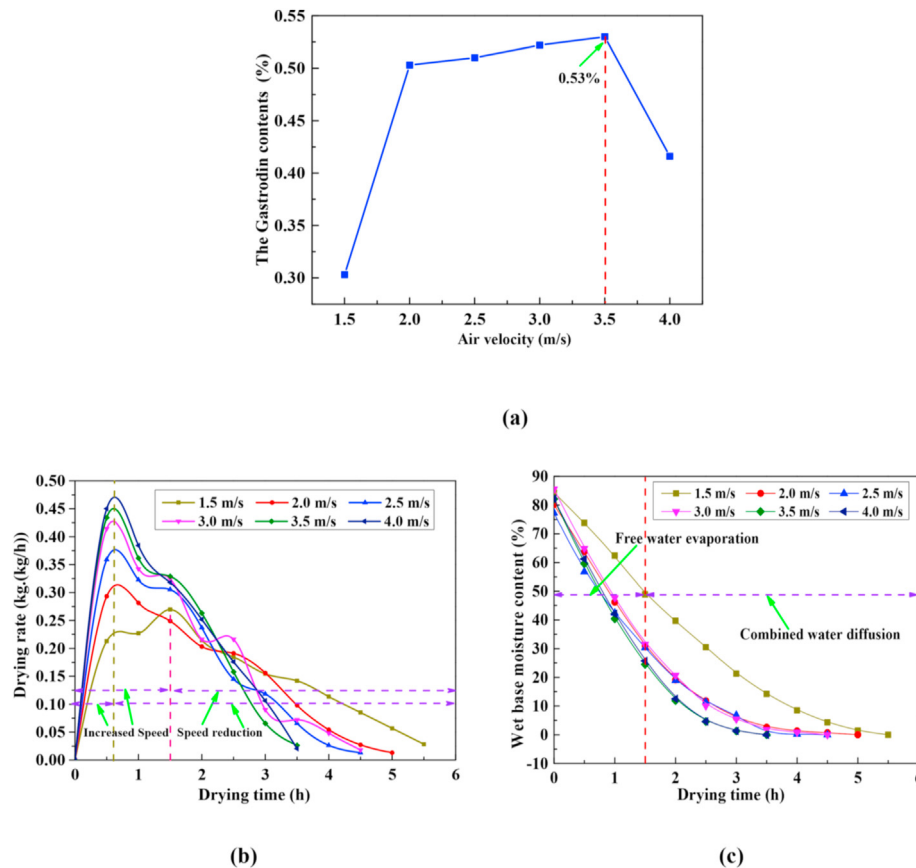


Fig. 4. Effect of air velocity on the: (a) gastrodin content; (b) drying rate; (c) water loss rate.

more gastrodin soluble in water. However, at high drying rates, gastrodin dissolved in water will also diffuse part of the gastrodin into the air as the internal boundary water diffuses. Therefore, the slice thickness has a significant influence on the gastrodin content. The gastrodin content at the slice thickness of 0.4 cm was 2.5 times higher than that of standard stipulated in the Chinese pharmacopoeia [8].

Moreover, it can be concluded that the drying rate is steeper with thinner slice thickness. The reason is that the thinner the gastrodia elata slices, the shorter the distance that the water in the material migrates to the surface during the evaporation process, and the faster the mass and heat exchange rate between the material and the air [27]. The pre-heating period takes 1.0 h at a slice thickness of 0.6 cm and 0.8 cm, while it takes a half-hour at a slice thickness of 0.2 cm and 0.4 cm with the same half of safe moisture content, as shown in Fig. 5(b).

Fig. 5(c) shows that with the increase of the slice thickness, the slope spacing between drying curves increases gradually. When the slice thickness was 0.2 cm–0.4 cm, the spacing between drying curves was the smallest. The water content of gastrodia elata decreases rapidly at the slice thickness of 0.4 cm. Therefore, slice thickness of gastrodia elata has a great influence on the drying process. The dried gastrodia elata has a crimp phenomenon at a slice thickness of 0.2 cm, but this phenomenon did not appear at a slice thickness of 0.4 cm. Therefore, gastrodia elata with a thickness of 0.4 cm is the optimum drying parameter, which gave the optimum drying rate and the highest content of gastrodin.

3.4. Effect of relative humidity on the characteristics of *gastrodia elata*

Fig. 6 shows the effect of relative humidity on the sliced gastrodia elata drying rate, water loss, and gastrodin content at constant slice thickness 0.4 cm, air velocity 3.5 m/s and temperature 60 °C in the hot air drying.

Fig. 6(a) depicts that as the relative air humidity increases from 20 % to 25 %, the gastrodin content increases from 0.53 % to 0.58 %. Whereas, as the relative humidity increases from 30 % to 35 %, the gastrodin content decreases rapidly from 0.557 % to 0.416 %. In addition, when the relative humidity was 25 %, the content of gastrodin was 2.9 higher than that of standard stipulated in the Chinese pharmacopoeia [8]. It can be concluded that low air relative humidity was beneficial to increase the activity of gastrodin generating enzymes and increase the conversion of gastrodia internal components to gastrodin. Besides, the gastrodin content is insensitive to the relative humidity.

Fig. 6(b) shows that the drying rate decreases gradually with the increase of relative humidity. The reason is that the lower the relative humidity of the air, the greater the vapour pressure difference between the gastrodia elata and the environment, the greater the driving force of heat transfer and mass transfer between the material and the air, and the higher the drying rate [28]. The pre-heating time was shorter at the relative humidity of 20 % and 25 % than that at the relative humidity of 30 % and 35 %. Therefore, when the relative humidity is less than 30%, the drying time is reduced. On the other hand, the slope of the drying rate curve is steeper with the lower relative humidity while the water content

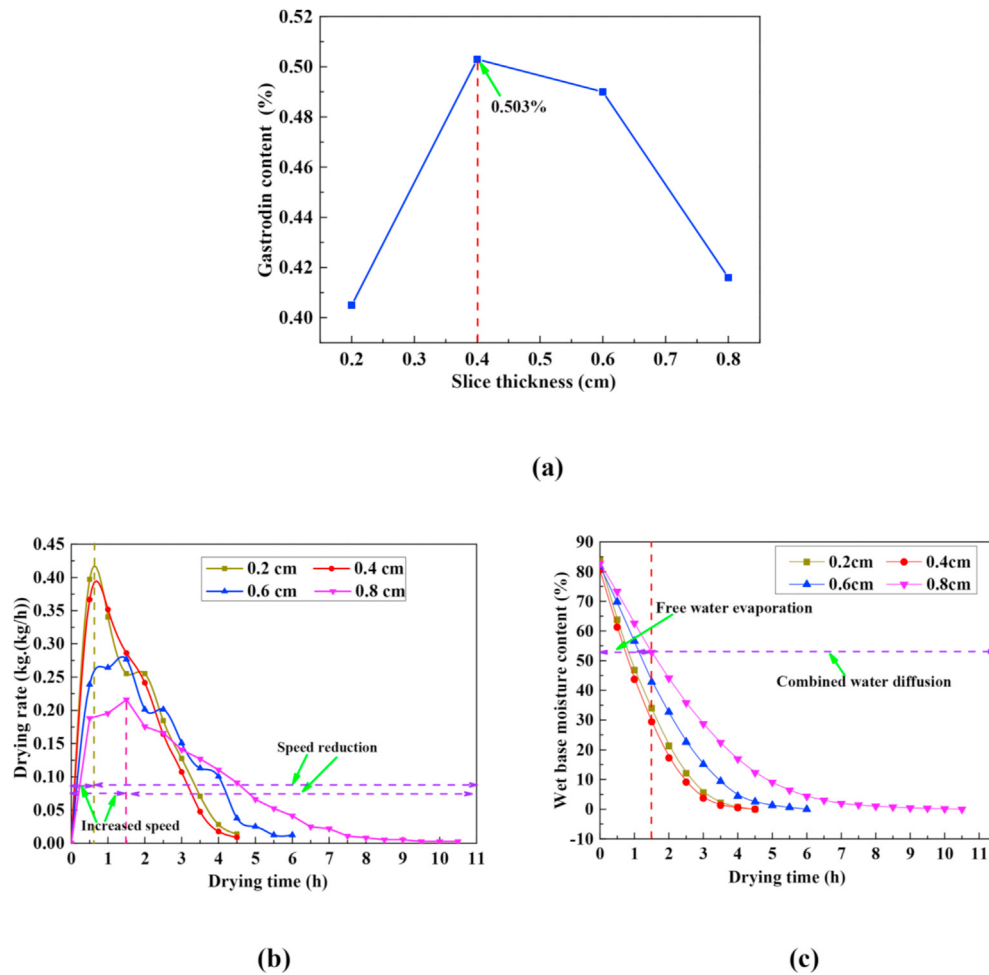


Fig. 5. Effect of slice thickness on the: (a) gastrodin content; (b) drying rate; (c) water loss rate.

drops faster, as shown in Fig. 6(c). However, slope spacing between the drying rate curves is not obvious with the relative humidity above 25 %, so the relative humidity has little effect on the water loss process of gastrodia elata. The optimum drying parameter of gastrodia elata at relative humidity is 25 %, which corresponds to the maximum content of gastrodin.

3.5. Dynamic models of drying gastrodia elata

Drying process of the material was a coupled process of heat and quality. On one hand, the wet material absorbs heat from the hot air. Then the heat was transferred from the outside of the wet material to the inside. On the other hand, the moisture inside the material was transferred from the inside to the outside until the moisture content of the material achieves to the drying requirements. The hydrodynamic model could predict the drying process of materials, which was established by analysing the Luvkov's equation and combining with Fick's law of diffusion. Drying constant and effective water diffusivity varies considerably within material due to the complex structure of the material and its interaction with water. Also, these parameters are strongly dependent on temperature and moisture content of the dried materials and air velocity [29,30]. By combining experiments with adequate modelling and computational simulation of heat and mass transfer between the material and the air, we can properly predict the performance of the drying material, as well as the variables that

affect the quality of the product [30].

Fick's law type equation was used to describe the phenomenon of heat migration and moisture migration in porous media under isothermal and non-isothermal conditions of liquid water and water vapour. Based on the theory of mass conservation, the irreversible law of thermodynamics is combined with Fick's law. Luvkov's equation regards the heat and mass exchange in the drying process of wet materials as a whole. This equation consists of three partial differential equations, which can describe the coupling of heat and mass transfer of wet materials in the drying process [29,32], as shown in Eqs. (5)–(7).

$$\frac{\partial T}{\partial t} = K_{11} \cdot \nabla^2 T + K_{12} \cdot \nabla^2 M + K_{13} \cdot \nabla^2 P \quad (5)$$

$$\frac{\partial M}{\partial t} = K_{21} \cdot \nabla^2 T + K_{22} \cdot \nabla^2 M + K_{23} \cdot \nabla^2 P \quad (6)$$

$$\frac{\partial P}{\partial t} = K_{31} \cdot \nabla^2 T + K_{32} \cdot \nabla^2 M + K_{33} \cdot \nabla^2 P \quad (7)$$

where P is the dry pressure, T is the dry temperature, M is the moisture content of materials, K_{11} – K_{23} is the coupling coefficient.

The pressure change has little effect on the heat transfer of the material during the drying process. Therefore, it's can ignore the influence of pressure. Assuming that the material temperature is evenly distributed and equal to the temperature of the drying

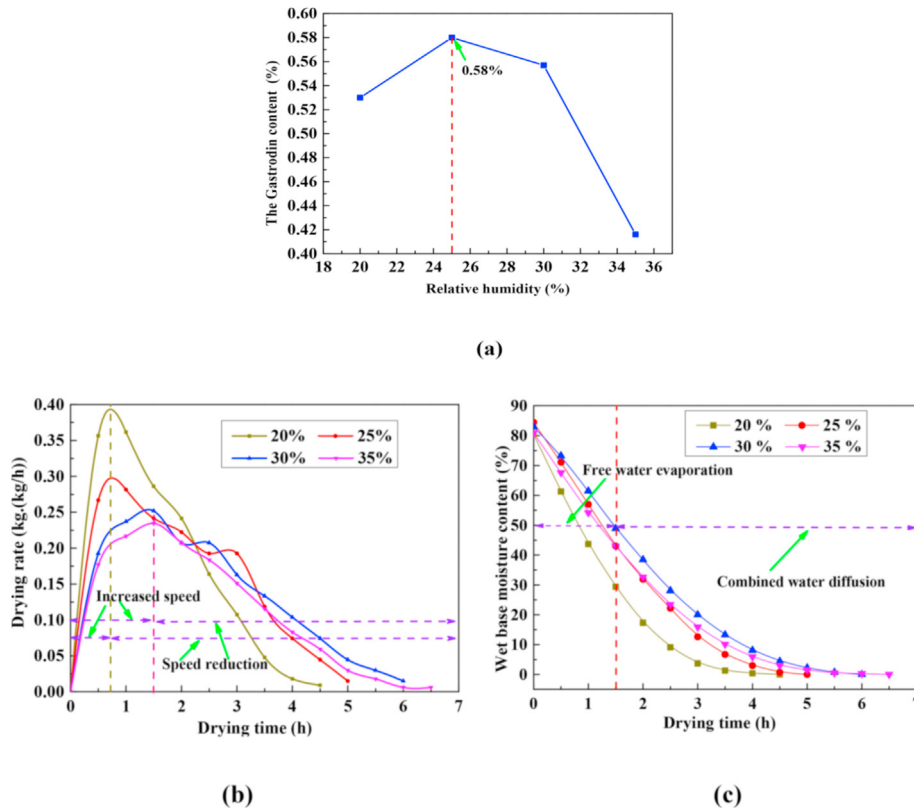


Fig. 6. Effect of air humidity on the: (a) gastrodin content; (b) drying rate; (c) water loss rate.

medium, the Luikov's partial differential equations are simplified as Eq (8) and Eq (9):

$$\frac{\partial T}{\partial t} = K_{11} \cdot \nabla^2 T \quad (8)$$

$$\frac{\partial M}{\partial t} = K_{22} \cdot \nabla^2 M \quad (9)$$

where K_{11} is the thermal diffusion coefficient u_n , K_{22} is the effective water diffusion coefficient (m^2/s).

The Luikov's equations can be changed to Eq (10), Eq (11).

$$\frac{\partial T}{\partial t} = u_n \cdot \left(\frac{\partial^2 T}{\partial x^2} + \frac{a_1}{x} \cdot \frac{\partial T}{\partial x} \right) \quad (10)$$

$$\frac{\partial M}{\partial t} = D_{eff} \cdot \left(\frac{\partial^2 M}{\partial x^2} + \frac{a_1}{x} \cdot \frac{\partial M}{\partial x} \right) \quad (11)$$

Due to the relatively uniform temperature distribution and heat conduction of the thin layer drying, the analytical formula of Eq 10 and 11 can be written as Eq (12).

$$MR = \frac{8}{\pi^2} \sum_{i=1}^{\infty} \frac{1}{(2i-1)^2} e^{\left(\frac{-(2i-1)^2 \pi^2 D_{eff} t}{4L^2} \right)} \quad (12)$$

Where the MR is the moisture ratio, D_{eff} is the effective water

diffusion coefficient (m^2/s), t is the drying time (s), L is the half-thickness of the slice (m).

For long drying time, only the first term of Eq (8) becomes relevant for evaluating MR [11,31]. Therefore, it is possible to simplify the previous expression as Eq (13):

$$MR = \frac{8}{\pi^2} e^{\left(-D_{eff} \left(\frac{\pi}{2L} \right)^2 t^n \right)} \quad (13)$$

It is worth mentioned that when $n = 1$, Eq (9) converges. The value of n can distinguish some of the domains of anomalous transport: sub-diffusion if $0 < n < 1$ or super-diffusion if $n > 1$. It should be mentioned that the generalized diffusion coefficient D_{eff} has the dimension of m^2/s^n . The threshold between both anomalous transport domains is normal Brownian diffusion [31].

Therefore, a general valid expression for Eq (13) is as follows Eq (14):

$$MR = k_1 e^{(k_2 t^n)} \quad (14)$$

where k_1 is constant and dependent on geometry (e.g., $k_1 = 0.81$ for an infinite slab), k_2 is a constant dependent on the D_{eff} and sample geometry, and n can take the value of 1 for Fick, or different values for anomalous cases.

The drying process could be characterized as Fickian ($n = 1$), anomalous sub-diffusive ($n < 1$) or anomalous super-diffusive ($n > 1$) where the differences on diffusion processes are mainly related with the food microstructure. Therefore, just one value of n should represent all the processes, independent of the drying

Table 3
Drying kinetics models.

Models	Model equations
Page	$MR = \exp(-kt^n)$
Modified Page	$MR = \exp[-(kt)^n]$
Logarithmic	$MR = a \exp(-kt) + b$
Henderson and Pabis	$MR = a \exp(-kt)$
Diffusion approximation	$MR = a \exp(-kt) + (1-a) \exp(-kbt)$
Two-term	$MR = a \exp(-kt) + b \exp(-k_1t)$
Newton	$MR = \exp(-kt)$

temperature (n constant), while k_2 must be increasing with temperature. Under these constraints, it is potential to fit all the drying data (considering the three temperatures) and search for a unique value of n and k_2 for each temperature [31].

Many scholars have derived a variety of kinetic models for predicting the water loss process of dry materials by analysing the Luikov equation and combining Fick's diffusion law. Seven models were reviewed and summarized in Table 3, which commonly used mathematical models [33–39]. Due to the physical and chemical differences between the dried materials, different drying models are suitable for different drying kinetic models. Therefore, the appropriate mathematical model should be clarified according to the experimental data of the dried materials.

The mathematical model of the gastrodia elata is selected with fitting and analysing the experimental data. The indicators used to evaluate the goodness of fit of the models are the correlation coefficient (R^2), chi-square value (χ^2) and root mean square error (RMSE). In this way, the optimal model fitting from the experimental data was identified with higher R^2 value, lower RMSE values and χ^2 values [40]. The validation metrics are calculated by Eq. 15–17.

$$R^2 = 1 - \frac{\sum_{i=1}^N (MR_{\text{exp},i} - MR_{\text{pre},i})^2}{\sum_{i=1}^N (MR_{\text{exp},i} - MR_{\text{pre},i})^2} \quad (15)$$

$$\chi^2 = \frac{\sum_{i=1}^N (MR_{\text{exp},i} - MR_{\text{pre},i})^2}{N - n_0} \quad (16)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (MR_{\text{exp},i} - MR_{\text{pre},i})^2}{N}} \quad (17)$$

3.6. Validation of drying kinetics model

The Table 4 shows errors obtained by the comparison between the experimental data of dry gastrodia elata and the seven mathematical models. The results showed the Page model has the highest accuracy, of which the indicator of R^2 were the largest, and the χ^2 , and RMSE were relatively smaller. The R^2 ranged from 99.6 % to 99.9 %, the χ^2 ranged from 1.92 % to 2.98 %, and RMSE ranged from 2.65 % to 4.94 %.

3.7. Optimized model establishment

The proposed equations were mounted in the experimental data by the direct least square method proposed by Davidson–Fletcher–Powell method. The least squares method is the

most commonly used method for curve or surface fitting. It is based on the minimum sum of squared errors as the criterion, according to the discrete data of the experiment, a function formula is estimated to fit these data, and the fitted curve can reflect the basic trend of the data.

The fitting of the equation was evaluated based on the goodness of fit by the coefficient of determination (R^2). The statistical results from parameters K and n of the Page model related to the four drying control variables were summarized in Table 5.

Table 4 shows that the drying parameter K of Page model was a polynomial function of the air velocity as shown in Eq. (19) and slice thickness as shown in Eq. (21), while it was a linear function of the temperature (Eq. (18)) and relative humidity (Eq. (20)). Accordingly, the Eq. (22) was developed to determine the drying parameter K of the Page model equation.

$$K = a_0 + a_1 T \quad (R^2 = 0.849) \quad (18)$$

$$K = a_0 + a_1 V + a_2 V^2 + a_3 V^3 \quad (R^2 = 0.992) \quad (19)$$

$$K = a_0 + a_1 U \quad (R^2 = 0.768) \quad (20)$$

$$K = a_0 + a_1 \phi + a_2 \phi^2 \quad (R^2 = 0.752) \quad (21)$$

$$K = -1.05074 + 0.01804T + 1.77764V - 0.45799V^2 + 0.03805V^3 - 0.68841U - 0.08627\phi + 0.00126\phi^2 \quad (R^2 = 0.857) \quad (22)$$

On the other hand, the drying parameter n of Page model was a polynomial function of the temperature (Eq. (23)), air velocity (Eq. (24)), relative humidity (Eq. (25)) and slice thickness (Eq. (26)). The Eq. (27) was developed to determine the drying parameter n of the Page model equation.

$$n = a'_0 + a'_1 T + a'_2 T^2 \quad (R^2 = 0.977) \quad (23)$$

$$n = a'_0 + a'_1 V + a'_2 V^2 + a'_3 V^3 \quad (R^2 = 0.990) \quad (24)$$

$$n = a'_0 + a'_1 U + a'_2 U^2 \quad (R^2 = 0.987) \quad (25)$$

$$n = a'_0 + a'_1 \phi + a'_2 \phi^2 \quad (R^2 = 0.881) \quad (26)$$

$$n = 1.85754 - 0.03572T + 5.11093 \times 10^{-4}T^2 - 0.19159V + 0.03005V^2 - 0.00156V^3 + 1.24418U - 1.25262U^2 - 0.01382\phi + 3.15546 \times 10^{-4}\phi^2 \quad (R^2 = 0.963) \quad (27)$$

The optimized Page model equation was developed as Eq. (28). This equation can be used to predict the water loss of gastrodia elata with the four drying variables.

Table 4
Model errors.

Drying condition	Model number	R^2	χ^2	RMSE
Temperature	1	0.996–0.998	4.11×10^{-4} – 1.34×10^{-3}	5.562×10^{-4} – 4.122×10^{-3}
	2	0.070–0.572	6.66×10^{-2} – 2.90×10^{-1}	1.888×10^{-1} – 4.662×10^{-1}
	3	0.993–0.997	2.83×10^{-4} – 2.11×10^{-3}	1.504×10^{-2} – 5.036×10^{-2}
	4	0.980–0.991	2.00×10^{-3} – 6.37×10^{-3}	4.000×10^{-2} – 6.911×10^{-2}
	5	5.077–6.272	7.66×10^{-2} – 1.39×10^{-1}	1.030×10^{-1} – 9.929×10^{-1}
	6	0.980–0.991	2.00×10^{-3} – 6.37×10^{-3}	3.996×10^{-2} – 6.911×10^{-2}
	7	0.974–0.991	2.1×10^{-3} – 7.53×10^{-3}	4.100×10^{-2} – 7.523×10^{-2}
Air velocity	1	0.996–0.999	1.92×10^{-4} – 2.98×10^{-3}	3.860×10^{-4} – 4.938×10^{-3}
	2	0	0	0
	3	0.994–2.440	7.96×10^{-4} – 2.14×10^{-1}	2.552×10^{-2} – 4.077×10^{-1}
	4	0.980–0.988	3.23×10^{-3} – 9.19×10^{-3}	5.010×10^{-2} – 8.103×10^{-2}
	5	0.980–0.988	0	0
	6	0.960–0.986	3.23×10^{-3} – 9.19×10^{-3}	5.010×10^{-2} – 8.103×10^{-2}
	7	0.960–0.986	3.65×10^{-3} – 1.05×10^{-2}	5.328×10^{-2} – 8.652×10^{-2}
Slice thickness	1	0.997–0.999	4.11×10^{-4} – 2.84×10^{-3}	2.651×10^{-4} – 4.866×10^{-3}
	2	0.970–0.981	1.84×10^{-3} – 5.94×10^{-3}	4.077×10^{-2} – 6.811×10^{-2}
	3	0.078–0.339	7.36×10^{-2} – 2.46×10^{-1}	2.581×10^{-1} – 4.376×10^{-1}
	4	0.000–0.994	6.29×10^{-4} – 2.08×10^{-3}	2.385×10^{-2} – 4.020×10^{-3}
	5	0.978–0.987	1.32×10^{-3} – 5.04×10^{-3}	3.453×10^{-2} – 6.260×10^{-2}
	6	5.200–6.535	3.41×10^{-2} – 1.32×10^{-1}	1.757×10^{-1} – 3.202×10^{-1}
	7	0.977–0.987	1.32×10^{-3} – 5.04×10^{-3}	3.453×10^{-2} – 6.260×10^{-2}
Relative humidity	1	0.997–0.998	4.11×10^{-4} – 1.56×10^{-3}	5.651×10^{-4} – 3.625×10^{-3}
	2	0	0	0
	3	0.990–0.994	1.02×10^{-3} – 2.08×10^{-3}	2.931×10^{-2} – 4.020×10^{-2}
	4	0.970–0.981	3.25×10^{-3} – 6.98×10^{-3}	5.246×10^{-2} – 7.469×10^{-2}
	5	0	0	0
	6	0.970–0.981	3.25×10^{-3} – 6.97×10^{-3}	5.246×10^{-2} – 7.469×10^{-2}
	7	0.963–0.978	4.11×10^{-3} – 8.68×10^{-3}	5.896×10^{-2} – 8.333×10^{-2}

Table 5
Parameters K and n of the Page model related to the four drying control variables.

Model parametric (K) equation	R^2	Model parametric (n) equation	R^2
$K = a_0 + a_1T + a_2T^2 + a_3T^3 + a_4T^4$	0	$n = a'_0 + a'_1T + a'_2T^2 + a'_3T^3 + a'_4T^4$	0
$K = a_0 + a_1T + a_2T^2 + a_3T^3$	0.742	$n = a'_0 + a'_1T + a'_2T^2 + a'_3T^3$	0.967
$K = a_0 + a_1T + a_2T^2$	0.841	$n = a'_0 + a'_1T + a'_2T^2$	0.977
$K = a_0 + a_1T$	0.849	$n = a'_0 + a'_1T$	0.972
$K = a_0 + a_1V + a_2V^2 + a_3V^3 + a_4V^4$	0	$n = a'_0 + a'_1V + a'_2V^2 + a'_3V^3 + a'_4V^4$	0.985
$K = a_0 + a_1V + a_2V^2 + a_3V^3$	0.992	$n = a'_0 + a'_1V + a'_2V^2 + a'_3V^3$	0.990
$K = a_0 + a_1V + a_2V^2$	0.917	$n = a'_0 + a'_1V + a'_2V^2$	0.977
$K = a_0 + a_1V$	0.928	$n = a'_0 + a'_1V$	0.921
$K = a_0 + a_1U + a_2U^2 + a_3U^3 + a_4U^4$	0	$n = a'_0 + a'_1U + a'_2U^2 + a'_3U^3 + a'_4U^4$	0
$K = a_0 + a_1U + a_2U^2 + a_3U^3$	0	$n = a'_0 + a'_1U + a'_2U^2 + a'_3U^3$	0
$K = a_0 + a_1U + a_2U^2$	0.690	$n = a'_0 + a'_1U + a'_2U^2$	0.987
$K = a_0 + a_1U$	0.768	$n = a'_0 + a'_1U$	0.500
$K = a_0 + a_1\phi + a_2\phi^2 + a_3\phi^3 + a_4\phi^4$	0	$n = a'_0 + a'_1\phi + a'_2\phi^2 + a'_3\phi^3 + a'_4\phi^4$	0
$K = a_0 + a_1\phi + a_2\phi^2 + a_3\phi^3$	0	$n = a'_0 + a'_1\phi + a'_2\phi^2 + a'_3\phi^3$	0
$K = a_0 + a_1\phi + a_2\phi^2$	0.752	$n = a'_0 + a'_1\phi + a'_2\phi^2$	0.881
$K = a_0 + a_1\phi$	0.714	$n = a'_0 + a'_1\phi$	0.832

$$MR = \exp \left[\left(+1.05074 - 0.01804T - 1.77764V + 0.45799V^2 - 0.03805V^3 + 0.68841U + 0.08627\phi \right. \right. \\ \left. \left. - 0.00126\phi^2 \right) t^{1.85754 - 0.03572T + 5.11093 \times 10^{-4}T^2 - 0.19159V + 0.03005V^2 - 0.00156V^3 + 1.24418U - 1.25262U^2 - 0.01382\phi + 3.15546 \times 10^{-4}\phi^2} \right] \quad (28)$$

3.8. Verification of the heat and moisture transfer model

Fig. 7 shows that the moisture ratio curve with the drying time under the different drying condition of slice *gastriodia elata* in the hot air drying. The line with the solid point stands for the predicted

data, and the line with the hollow point stands for the actual data.

Fig. 7 shows that the predicted MR curve is consistent with the actual curve with the different drying process parameters. The dot-solid line represents the actual value, while the dot-dash line represents the predicted value, where the actual values were under the

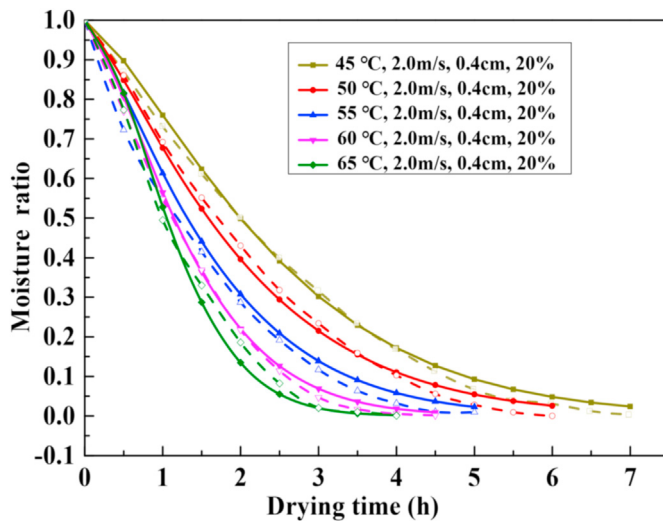


Fig. 7. The hydrodynamics model verification.

prediction curve. The Eq. (18) was applied to calculate the experimental MR results (individual drying samples with different drying process parameters), where the derived R^2 ranges between 99.37 % and 99.87 %, the $RMSE$ ranges between 1.27 % and 2.90 %, and the χ^2 ranges between 0.13 % and 0.59 %. It indicates that the optimized Page model equation can predict the water loss process of the *gastrodia elata* at different drying process parameters.

4. Calculation of effective water diffusion coefficient and activation energy

4.1. Effective water diffusion coefficient

The drying process of material was affected by the change of moisture distribution and content in the material, and the water diffusion coefficient was usually used to describe the mass transfer performance of water in a material [29]. The moisture effective diffusivity (D_{eff}) is generally considered as an important transport indicator of drying hydrodynamic, the large moisture effective diffusivity, the higher the diffusion rate of water molecules inside the material [41]. Assumed to neglect the shrinkage, uniform initial sample moisture content distribution, constant heat, and mass transfer diffusivity, the solution of the Fick's second law of diffusion can be given for various sample geometrics during the falling drying rate period. On this basis, the water diffusion coefficient was described by Fick's second law [30], the solution of the Fick diffusion equation can be simplified to Eq. (29).

$$MR = \frac{8}{\pi^2} \exp\left(-\frac{\pi^2 D_{eff} t}{4L^2}\right) \quad (29)$$

where the *gastrodia elata* slice sample is considered as the homogeneous slab, D_{eff} is the effective moisture diffusivity (m^2/s), and L is the half thickness of the slab (m).

Taking the logarithm of Eq. (29) on both sides to obtain the Eq. (30).

$$\ln MR = \ln \frac{8}{\pi^2} - \frac{\pi^2 D_{eff} t}{L^2} \quad (30)$$

According to Eq. (30), $\ln MR$ is the y-coordinate and t is the x-coordinate, it is clear that the plot of $\ln MR$ versus t (min) would be linear, and the slope can be used to calculate the effective moisture

diffusion coefficient (D_{eff}).

4.2. Calculation of activation energy

Activation energy was energy required to remove a unit of water during the drying process. The energy of drying material increased with the increase of the activation energy [31]. Based on the relationship between the effective diffusion coefficient (D_{eff}) in Fick's second Law and the hot air temperature T during drying, the activation energy can be described by an Arrhenius type relation [32], as shown in Eq. (31).

$$D_{eff} = D_0 \exp\left(\frac{E_a}{RT}\right) \quad (31)$$

where the E_a is the activation energy (kJ/mol), R is the universal gas constant (kJ/mol K), D_0 is the pre-exponential parameter (m^2/s) and T is the temperature (K).

Taking the logarithm of both sides of Eq. (21) to obtain Eq. (32):

$$-\ln D_{eff} = -\ln D_0 + \frac{E_a}{RT} \quad (32)$$

According to Eq. (32), the plot of $-\ln(D_{eff})$ versus $1/T$ (K) would be linear, and the slope can be used to calculate E_a/R [26,27].

4.3. Calculation of effective diffusion coefficient on *gastrodia elata* slices

The effective diffusion coefficient of the sliced *gastrodia elata* increased from $7.510 \times 10^{-9} m^2/s$ to $1.654 \times 10^{-8} m^2/s$ with the increase of the temperature (45 °C–60 °C), air velocity (2.0 m/s – 4.0 m/s) in the hot air drying. However, it decreased from $3.241 \times 10^{-8} m^2/s$ to $2.254 \times 10^{-9} m^2/s$ with the increase of the slice thickness (0.2 cm–0.8 cm), relative humidity (20%–35%), as shown in Table 6 and Table 7. This indicates that the shrinkage behaviour reduces the water diffusion rate, which is consistent with the result of drying rate decreasing with the extension time. However, the effective diffusion coefficient decreased with the increase of the slice thickness and relative humidity, because the relative humidity of the air reduced and the vapour pressure difference increased with the decrease of the slice thickness and relative humidity, which accelerated the mass transfer rate.

4.4. Calculation of activation energy on *gastrodia elata* slices

Fig. 8 shows a linear relationship between the effective diffusion coefficient and temperature. When the temperature decreased from 338 K to 318 K the effective diffusion coefficient increases from 18.013 kJ/mol to 18.707 kJ/mol. The value of E_a/R is equal to 4108.96 and the value of $-\ln D_0$ is equal to 5.806 m^2/s . Thus, the activation energy is 34.16 kJ/mol, which is similar to Chinese herbal medicines as the *Codonopsis* and *Poria Cocos* [42,43]. However, the activation energy of *gastrodia* is higher than potato chips (22.98 kJ/mol) and rapeseed (28.47 kJ/mol) [44,45]. This indicates that the *gastrodia elata* is more difficult to be dried, which requires more energy.

Table 6

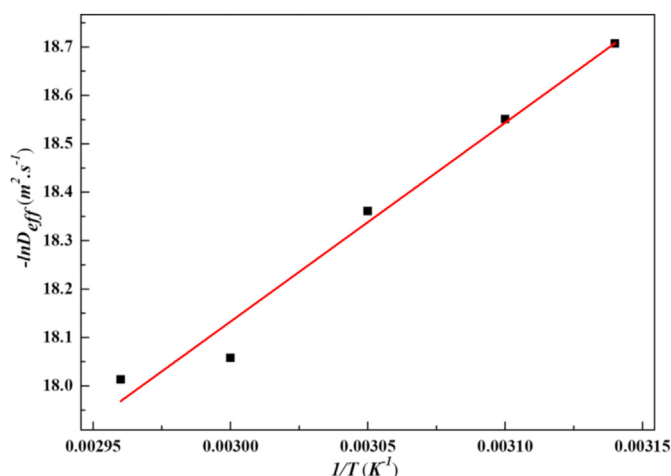
The range of effective water diffusivity under different drying conditions.

Drying condition	D_{eff} (m^2/s)
Temperature	7.510×10^{-9} – 1.504×10^{-8}
Air velocity	8.376×10^{-9} – 1.654×10^{-8}
Slice thickness	2.254×10^{-9} – 3.241×10^{-8}
Relative humidity	9.000×10^{-9} – 1.437×10^{-8}

Table 7

Moisture effective diffusion coefficient of gastrodia elata tuber slices under different drying conditions.

	Temperature °C	Air velocity m/s	Slice thickness cm	Relative humidity %	Linear equation expression	R ²	D _{eff} ² m/s
Temperature	45	2.0	0.4	20	$\ln MR = -1.85292 \times 10^{-4}t + 0.49193$	0.938	7.510×10^{-9}
	50	2.0	0.4	20	$\ln MR = -2.16555 \times 10^{-4}t + 0.49413$	0.927	8.777×10^{-9}
	55	2.0	0.4	20	$\ln MR = -2.61753 \times 10^{-4}t + 0.36163$	0.949	1.061×10^{-8}
	60	2.0	0.4	20	$\ln MR = 3.54522 \times 10^{-4}t + 0.56517$	0.946	1.437×10^{-8}
	65	2.0	0.4	20	$\ln MR = -3.71001 \times 10^{-4}t + 0.49911$	0.945	1.504×10^{-8}
Air velocity	60	1.5	0.4	20	$\ln MR = -2.06664 \times 10^{-4}t + 0.47054$	0.909	8.376×10^{-9}
	60	2.0	0.4	20	$\ln MR = -3.54522 \times 10^{-4}t + 0.56517$	0.946	1.437×10^{-8}
	60	2.5	0.4	20	$\ln MR = 3.71191 \times 10^{-4}t + 0.40911$	0.943	1.504×10^{-8}
	60	3.0	0.4	20	$\ln MR = 3.72888 \times 10^{-4}t + 0.43762$	0.937	1.511×10^{-8}
	60	3.5	0.4	20	$\ln MR = -3.7375 \times 10^{-4}t + 0.53035$	0.965	1.515×10^{-8}
Slice thickness	60	4.0	0.4	20	$\ln MR = -4.08058 \times 10^{-4}t + 0.49368$	0.978	1.654×10^{-8}
	60	2.0	0.2	20	$\ln MR = -3.19856 \times 10^{-4}t + 0.50389$	0.938	3.241×10^{-8}
	60	2.0	0.4	20	$\ln MR = -3.54522 \times 10^{-4}t + 0.56517$	0.946	1.437×10^{-8}
	60	2.0	0.6	20	$\ln MR = -2.46661 \times 10^{-4}t + 0.55353$	0.954	2.849×10^{-9}
	60	2.0	0.8	20	$\ln MR = -1.76078 \times 10^{-4}t + 0.61206$	0.976	2.254×10^{-9}
Relative humidity	60	2.0	0.4	20	$\ln MR = -3.54522 \times 10^{-4}t + 0.56517$	0.946	1.437×10^{-8}
	60	2.0	0.4	25	$\ln MR = -2.68011 \times 10^{-4}t + 0.56783$	0.905	1.086×10^{-8}
	60	2.0	0.4	30	$\ln MR = -2.56309 \times 10^{-4}t + 0.65824$	0.943	1.039×10^{-8}
	60	2.0	0.4	35	$\ln MR = -2.22054 \times 10^{-4}t + 0.55624$	0.924	9.000×10^{-9}

**Fig. 8.** Relationship between effective diffusion coefficient and temperature.

5. Conclusions

In this study, a comprehensive analysis is conducted on the influence of hot air drying parameters of gastrodia elata with the medicinal indicator of gastrodin ingredient to evaluate the drying quality. Meanwhile, established an optimized dynamic model with multiple drying variables for drying the sliced gastrodia elata with hot air. The conclusions are summarized as following:

- (1). Comparison of seven drying methods showed that Page model was the better to predict water loss process of gastrodia elata in the hot air dryer. The optimized Page model equation can accurately predict the drying dehydration process of the gastrodia elata under drying variables of temperature, air velocity, slice thickness.
- (2). In the hot air drying of the gastrodia elata, the temperature, and slice thickness was the main factors on the drying water

loss process of the gastrodia elata. Furthermore, the temperature and air velocity were the main factors on the increase of gastrodin content.

- (3). The optimum parameters to obtain the maximum content of gastrodin with short drying time in a hot air drying corresponded to the temperature of 60 °C, air velocity of 3.5 m/s, slice thickness of 0.4 cm, and the relative humidity of 25 %.
- (4). The effective diffusion coefficient of gastrodia elata in the hot air drying ranged from $2.254 \times 10^{-9} \text{ m}^2/\text{s}$ to $3.241 \times 10^{-8} \text{ m}^2/\text{s}$, which increased with increasing the temperature and air velocity and decreased with increasing the slice thickness and relative humidity. Meanwhile, the drying activation energy of gastrodia elata was 34.16 kJ/mol.

Credit author statement

Li Kun: Investigation, Data curation, Validation, Formal analysis, Writing – original draft preparation. Li Ming: Supervision, Project administration. Wang Yun Feng: Methodology, Software. Zhang Ying: Writing- Reviewing and Editing. Wael El-Kolaly: Writing- Reviewing and Editing. Gao Meng: Data curation, Visualization. Sun Wei: Software, Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

D_s	Drying rate (kg/(kg.h))
t	Drying time (h)
C_t	Moisture quality of wet materials (kg)
W_g	Absolutely dry material quality (kg)
$M_{t+\Delta t}$	Moisture content at time $t+\Delta t$ (%)
M_0	Equilibrium moisture content (%)
M_c	Wet base moisture content (%)
M_t	Dry basis moisture content at time t (%)
MR	Moisture ratio
D_{eff}	Effective diffusion coefficient (m^2/s)
E_a	Activation energy (KJ/mol)
D_0	Constant
L	One half of the thickness of dry material (m)
R	Gas constant
$MR_{exp, i}$	Experimental moisture ratio found in any measurement
$MR_{pre, i}$	Predicted moisture ratio obtained by calculating from the model
N	Number of observation
n_0	Number of model constants
R^2	Correlation coefficient
χ^2	Chi-square value
$RMSE$	Root mean square error
K	Drying model parameter
n	Drying model parameter
T	Drying air temperature ($^{\circ}C$)
V	Drying air velocity (m/s)
U	Slice thickness (cm)
Φ	Relative humidity of drying air (%)
a_0, a_1, a_2, a_3, a_4	Parameter of equation K
$a'_0, a'_1, a'_2, a'_3, a'_4$	Parameter of equation n

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